

EXPERIMENT 3

1.Aim:

To design and develop a Library Book Issue Management application using Zoho Creator, a low-code application development platform. The application is designed to digitize the traditional library book issue request process by allowing students, faculty, and staff to submit book issue requests through an online form. The application uses all 13 Basic Field types available in the Zoho Creator Form Builder to collect member details, contact information, book requirements, preferred collection schedule, and acknowledgement of library rules. The system also provides a report view for library staff to verify requests, check book availability, filter requests, and process book issue requirements efficiently.

2. Traditional Application Development vs Emerging Trends in Low-Code Application Development

Traditional application development is a conventional method of building software applications using programming languages, frameworks, databases, and development tools. In this approach, developers manually write code for different parts of the application, including the user interface, database operations, validation, business logic, and security. The development process usually involves several stages such as requirement analysis, design, coding, testing, deployment, and maintenance. Although traditional development provides greater control and customization, it generally requires more programming knowledge, development time, and technical resources.

Low-code application development is an emerging approach that simplifies the software development process by providing visual development environments, drag-and-drop components, pre-built fields, workflows, and automation features. Platforms such as Zoho Creator allow users to develop applications with minimal manual coding. For example, in the Library Book Issue Management application, fields such as Name, Email, Phone, Address, Number, Date, Time, Drop Down, Radio Button, Multi Select, and Checkbox can be added directly using the Form Builder. Required validations, reports, and workflows can also be configured through the platform. This reduces development time and makes application development accessible to users with limited programming knowledge.

The major difference is that traditional development depends heavily on manual programming, while low-code development focuses on visual configuration and reusable components. Traditional development is more suitable when an application requires highly customized functionality and complete control over the underlying system, whereas low-code development is useful for quickly developing business applications, forms, workflows, and prototypes. Therefore, low-code development represents an emerging trend that helps organizations develop and modify applications more quickly while reducing the amount of manual coding required.

3. Benefits and Challenges in Low Code.

3.1 Benefits of Low-Code Development

1. Faster Development

Low-code platforms provide pre-built components and drag-and-drop functionality, reducing the time required to develop an application.

2. Less Programming

Developers do not need to manually write code for every component. Many features can be configured visually.

3. Easy to Use

The visual interface makes low-code platforms easier to understand for beginners and users with limited programming experience.

4. Rapid Prototyping

A working prototype can be created quickly. This helps organizations test an idea before developing a large-scale system.

5. Easy Form Creation

Forms can be created using predefined fields such as text, email, phone, date, time, dropdown, radio buttons, and checkboxes.

6. Workflow Automation

Tasks such as email notifications, record processing, approvals, and reminders can be automated.

7. Reduced Development Cost

Since less manual coding and development time are required, the overall development cost can be reduced.

8. Easy Maintenance

Changes can often be made through the visual interface without modifying large amounts of source code.

9. Better Productivity

Developers and business users can concentrate on application requirements instead of spending time implementing basic features.

10. Cloud Accessibility

Applications created using cloud-based low-code platforms can generally be accessed from different devices and locations.

3.2 Challenges of Low-Code Development

1. Limited Customization

Highly complex requirements may not be easily implemented using only the available visual components.

2. Platform Dependency

Applications may become dependent on the selected low-code platform and its features.

3. Security Concerns

Sensitive organizational data must be properly protected using appropriate authentication, authorization, and security configurations.

4. Performance Limitations

For very large or highly complex applications, platform limitations may affect performance.

5. Vendor Lock-In

Moving an application from one low-code platform to another may require significant changes.

6. Limited Control

Developers may have less control over the underlying infrastructure and implementation compared with traditional development.

7. Learning Platform-Specific Features

Although less programming is required, users still need to understand the specific platform, its features, workflows, and configuration options.

8. Integration Complexity

Connecting with complex external systems may still require APIs, custom scripts, or programming knowledge.

4. COMPARISON OF EXISTING LOW-CODE APPLICATION DEVELOPMENT PLATFORM

The following table compares some commonly used low-code application development platforms.

Feature	Zoho Creator	Microsoft Power Apps	Google AppSheet	Salesforce Platform
Development Type	Low-code	Low-code	No-code/Low-code	Low-code
Form Creation	Available	Available	Available	Available

Drag-and-Drop Interface	Yes	Yes	Yes	Yes
Workflow Automation	Yes	Yes	Yes	Yes
Database/Data Management	Available	Available	Available	Available
Report/Dashboard Features	Yes	Yes	Yes	Yes
Email Notifications	Supported	Supported	Supported	Supported
Integration	Supports APIs and external services	Strong Microsoft ecosystem integration	Strong Google ecosystem integration	Strong CRM/business integration
Suitable Applications	Business forms, workflows, management applications	Business applications and automation	Data-driven business applications	CRM and enterprise applications

5. Result

The Library Book Issue Management application was successfully developed using Zoho Creator. The form uses all 13 Basic Fields with the required validations and allows members to submit book requests digitally. A report was also created to help library staff view, filter, and process the submitted requests efficiently.